

Supervisor Feedback on Proposal

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Title: Design and Development of a Web-Based System to Support Online Learning for Students at NUST

1. Topic Suitability

The topic addresses an important and relevant issue. However, in its current form, it is too broad and lacks the technical depth required for an Honours degree in Software Engineering.

You must refine the scope to focus on a specific, well-defined problem within online learning.

2. Problem Definition

The current problem statement is too general and reads more like an educational challenge rather than a software engineering problem.

You are required to:

- Clearly define a specific technical problem
- Avoid broad statements such as “students struggle with online learning”
- Identify a precise gap that can be addressed through system design

3. Risk of Rebuilding Existing Systems

The proposal currently risks duplicating functionality already provided by existing platforms.

You must clearly articulate:

- What existing systems do not solve effectively
- Why a new system is necessary
- What unique capability your solution introduces

4. Technical Depth (Critical)

The proposal lacks sufficient technical depth.

At Honours level, you are expected to demonstrate:

- System architecture design

- Algorithmic or logic-based solutions
- Data-driven, analytical, or optimization components where applicable

Simply developing a web-based system is not sufficient.

5. Research Contribution

You must clearly define:

- What new knowledge or insight your study contributes
- How your system advances current approaches

At present, the proposal does not demonstrate a clear research contribution.

6. Evaluation Strategy

The evaluation approach is currently too limited.

You must include:

- Measurable evaluation criteria (e.g., performance, efficiency, engagement)
- Comparison with existing tools or baseline systems
- Justification of how effectiveness will be assessed

7. Scope Refinement Required

You are required to significantly narrow your focus. Examples of acceptable directions include:

- Low-bandwidth optimized learning systems
- Intelligent content delivery systems
- Student engagement tracking and analytics
- Adaptive or personalized learning support

8. Additional Considerations

Including lecturers' perspectives can strengthen the study, but only if it supports a clearly defined technical solution.

Exploring whether challenges originate from online or on-campus learning is more aligned with educational research and should only be included if it directly informs system design.

9. Supervisor Expectation

This project must be treated as a software engineering and research project.

You are expected to:

- Demonstrate continuous progress
- Develop and present working prototypes throughout the semester
- Apply appropriate engineering and research methodologies

You are not simply building an application; you are developing a solution supported by technical design and evaluation.

Final Comment

The topic has potential but requires major refinement to meet Honours-level expectations in Software Engineering.

Approval is therefore conditional upon:

- Clear problem definition
- Strong technical contribution
- Well-defined system scope
- Measurable evaluation framework